

```
import pandas as pd
from sklearn.linear_model import LinearRegression

data_multi = {
    'area': [2600, 3000, 3200, 3600, 4000],
    'bedrooms': [3, 4, np.nan, 3, 5],
    'age': [20, 15, 18, 30, 8],
    'price': [550000, 565000, 610000, 595000, 760000]
}
df2 = pd.DataFrame(data_multi)

print("--- TASK 2: MULTIVARIATE LINEAR REGRESSION ---")
median_bedrooms = df2.bedrooms.median()
df2.bedrooms = df2.bedrooms.fillna(median_bedrooms)
print("Data after handling missing values:")
print(df2)

X_multi = df2[['area', 'bedrooms', 'age']]
y_multi = df2['price']

reg_multi = LinearRegression()
reg_multi.fit(X_multi, y_multi)

print(f"\nCoefficients [area, bedrooms, age]: {reg_multi.coef_}")
print(f"Intercept: {reg_multi.intercept_:.2f}")

predicted_custom = reg_multi.predict(pd.DataFrame([[3000, 3, 40]], columns=['area', 'bedrooms', 'age']))
print(f"Predicted Price for custom house: ${predicted_custom[0]:.2f}")
```

```
--- TASK 2: MULTIVARIATE LINEAR REGRESSION ---
Data after handling missing values:
```

	area	bedrooms	age	price
0	2600	3.0	20	550000
1	3000	4.0	15	565000
2	3200	3.5	18	610000
3	3600	3.0	30	595000
4	4000	5.0	8	760000

```
Coefficients [area, bedrooms, age]: [ 198.47159002 -116583.73881651 -14267.77585936]
Intercept: 656046.54
Predicted Price for custom house: $330999.06
```